Investment and Financial Management

Financial Markets

Stocks

Stocks – Legal definition

- The share capital (ownership rights) of a corporation is divided into single securitized shares representing the equity claim of the corresponding shareholder.
- In Germany, we distinguish between share capital (Grundkapital) and capital reserves (Kapitalrücklage). The share capital is equal to the face value of a stock multiplied with the number of stocks issued (minimum face value is 1 Euro). The capital reserves are build by the agio (difference to face value) paid by investors at the time when shares are issued.
- In other jurisdictions, like the US, this distinction is not made. Economically it is irrelevant.

Stocks – Symbols and terminology

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Symbol	Meaning
P_0	$Share\ price\ at\ t_0$
D_t	$Dividend\ per\ share\ in\ period\ (year)\ t$
EPS_t	$Earnings\ per\ Share\ in\ period\ (year)\ t$
r_e	$Costs\ of\ equity$
$\frac{P}{E}$	$Price - earnings - ratio$
$\frac{P}{B}$	$Price - book\ value - ratio$
w	$growth\ rate$
p	$dividend\ ratio$
ROE	$Return\ on\ equity$
V_e	$Equity\ value$

Stocks – Symbols and terminology

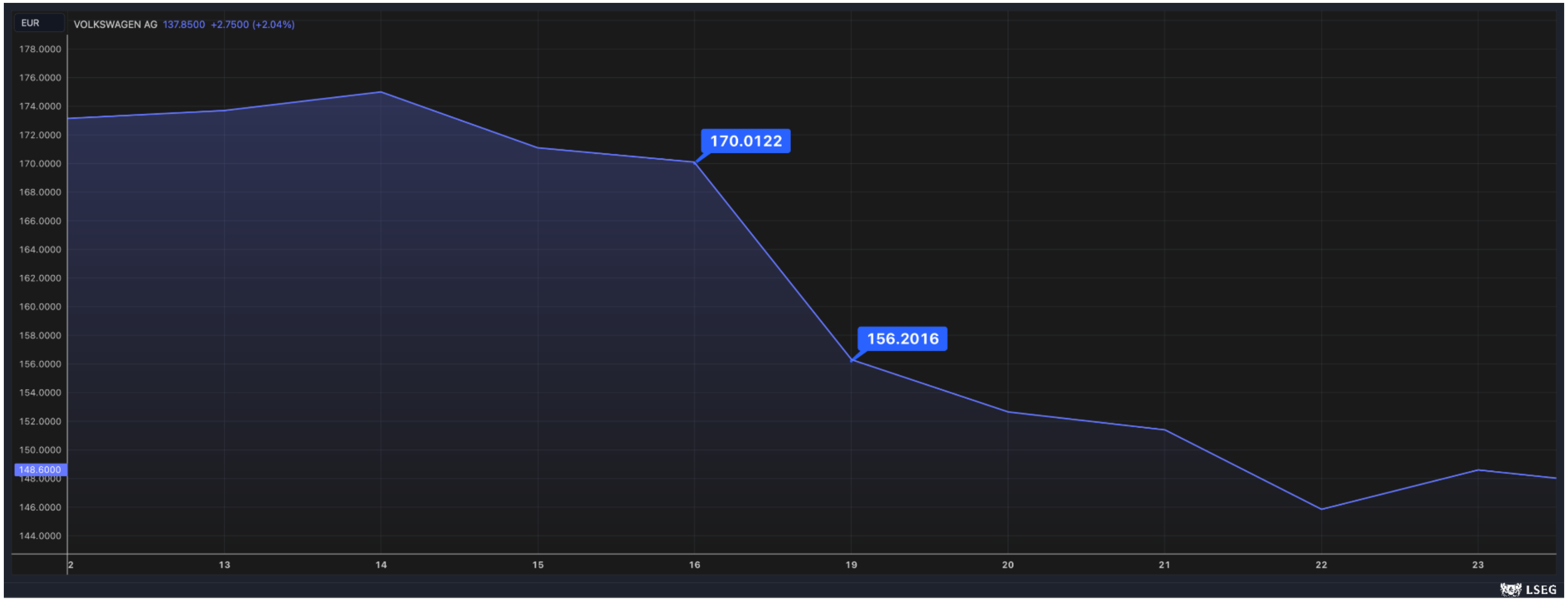
Dividends are the proportion of a company's earnings which is distributed to the shareholders

- Two types of dividends
  - Cash dividend
  - Stock dividend (e.g. Deutsche Telekom 2013)

- Dividends are paid regularly
  - Yearly in Germany
  - Quarterly in the United States

- Paying dividends is not mandatory but optional

Stock Price Reaction to Dividend Payments
VW ordinary shares' price (12.12.2022 - 23.12.2022)



- On December 19, 2022, Volkswagen shareholders were entitled to a special dividend of 19.06 € (effective payout was on January 9th, 2023).
- On that day the stock opened at 152.80 €; therefore, the stock price fell by (170.1-152.80=) 17.30 €.
- On the same day the DAX was almost flat (+0.30%).

* Special dividend of € 33 and regular dividend of € 1.80

Dividends of German DAX corporations

Company	Ann. Date	Ex-Date	Dividend 2026 (EUR)	Price (EUR)	Dividend yield (%)
Adidas	May 4, 2026	May 8, 2026	2.80	141.70	1.98%
Airbus	Feb 20, 2026	Apr 21, 2026	3.20	176.38	1.81%
Allianz	Feb 26, 2026	May 8, 2026	17.10	388.70	4.40%
BASF	Mar 4, 2026	May 4, 2026	2.25	54.74	4.11%
Bayer	Mar 4, 2026	Apr 27, 2026	0.11	38.50	0.29%

Dividend ratios

Dividends are the proportion of a company's earnings which is distributed to the shareholders

Dividend per share

$$(1): Dividend\ per\ share = \frac{Sum\ of\ all\ dividend\ payments}{Number\ of\ shares\ outstanding}$$

Dividend yield

$$(2): Dividend\ yield = \frac{Dividend\ per\ share}{Share\ price}$$

Payout ratio (proportion of the firm's earnings that is distributed)

$$(3): Payout\ ratio\ (dividend\ ratio) = \frac{Dividend\ per\ share}{Earnings\ per\ share}$$

Dividend ratios

(A.1) For the fiscal year 2016 a dividend was paid on May 12, 2017. The sum of all dividend payments was equal to 3,047 million EUR. The number of shares outstanding was 4,358 million. Out of these shares 5 million shares have been held by the firm itself. The earnings per share (EPS) for 2016 were 1.00 EUR. The stock was priced 9.165 EUR on May 12, 2017.

Dividend per share

$$(1)\!: Dividend\ per\ share = \frac{Sum\ of\ all\ dividend\ payments}{Number\ of\ shares\ outstanding} = \frac{3{,}047}{4{,}358 - 5} = 0.70\ EUR$$

Dividend yield

$$(2)\!: Dividend\ yield = \frac{Dividend\ per\ share}{Share\ price} = \frac{0.70}{9.165} = 7.64\%$$

Payout ratio:

$$(3)\!: Payout\ ratio\ (dividend\ ratio) = \frac{Dividend\ per\ share}{Earnings\ per\ share} = \frac{0.70}{1.00} = 70\%$$

Overview

Pricing of stocks

"Price is what you pay, value is what you get"

<u>Law of one price</u>: The price of a stock should equal <u>the present value of its future cash flows</u> distributed to the shareholder (**discounted cash flow principle**).

- In Reality: The **value of a stock can differ from its price** which is the result of specific conditions within a specific market environment with non-efficient market mechanisms.
- **Thus**: We assume a **perfect capital market** so that the price of a stock equals its fundamental value.
- **Necessary conditions** for perfect capital market:
  - Large number of investors without market power
  - Equal access to information by all market participants
  - Completely rational economic actors
  - No transaction costs (e.g. taxes)

Efficient Market Hypothesis

Do market prices reflect fundamental values?

Eugene Fama

“*In an efficient market, at any point in time, the actual price of a security will be a good estimate of its intrinsic value.*”

VS

Richard Thaler

“*I don’t think anyone thinks that the value of the world economy fell 25 percent that day. Nothing happened.*” Referring to a stock market crash on Monday 19, October 1987.

Overview

Pricing of stocks

The price of a stock should equal the present value of its future cash flows distributed to the shareholder

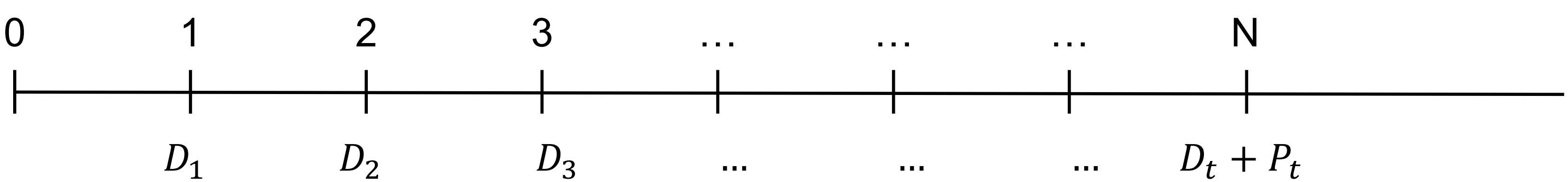
Similar to a bond (coupon payments and payments through the sale of the security), a stock offers two sources of payments…

1. Dividend payments D_t
2. Payments through the sale of the share with selling price P_t

But: Future dividend payments and future selling price are **uncertain** and not easy to predict.

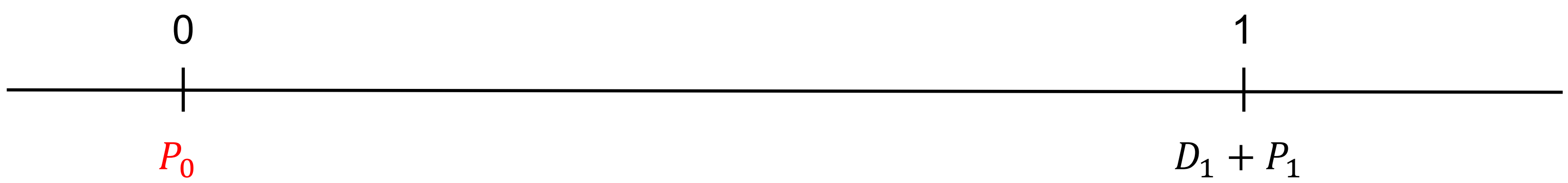
<u>Assumption due to simplification:</u>

- Only one dividend payment at the end of each year



Stock return with a one-year investment horizon

I. Investor has a one-year investment horizon
- Investor buys the share at the current price: P_0
- When holding the share the investor has the right to receive dividends D_1
- At the end of the year the investor sells the share at the share price P_1



The investor's return will be equal…

$$(1): r_e = \frac{D_1 + P_1}{P_0} - 1 = \underbrace{\frac{D_1}{P_0}}_{\text{Dividend yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Price gains or losses}}$$

… to the sum of dividend yield and relative price gains or losses.

Stock return with a one-year investment horizon

- Based on the investor's expectations about D_1, P_1, and as a consequence about r_e the investor is willing to pay a certain price for the share. Given this price, the net present value will be equal to zero.
- Due to the uncertainty of the cash flows they need to be **discounted with the costs of equity** (risk-adjusted interest rate, expected return r_e).
- The **costs of equity represent the expected return** which would be derived from comparable investments with a similar risk profile (equivalent to discount rate r in the bond pricing environment).

0 1

$$P_0 \xleftarrow{(D_1+P_1) * (1+r_e)^{-1}} D_1 + P_1$$

$$(1): r_e = \frac{D_1 + P_1}{P_0} - 1 = \frac{D_1}{P_0} + \frac{P_1 - P_0}{P_0}$$

$$(2): P_0 = \frac{D_1 + P_1}{1 + r_e}$$

Stock return with a one-year investment horizon

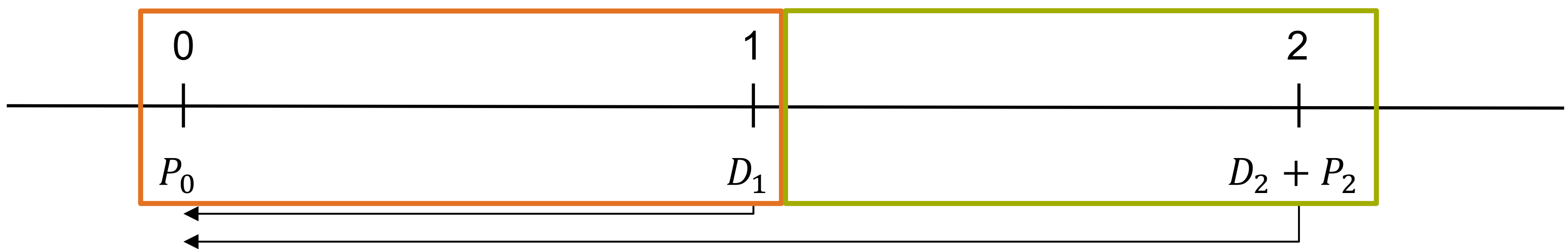
(A.2) An investor expects the corporation Long Drug Store to pay a dividend of $0.56\ EUR$ per share in the following year. Her expectations predict the shares to be traded at a price of $45.50\ EUR$. The return of comparable investments with a similar risk profile equal 6.8%. Please compute the maximum price the investor would be willing to pay for the shares today.

$$(1): P_0 = \frac{D_1 + P_1}{1 + r_e} = \frac{0.56 + 45.50}{1.068} = 43.13\ EUR$$

Stock return with a two-year investment horizon

II. Investor has a two-year investment horizon

- Investor buys the share at the current price: P_0
- When holding the share the investor has the right to receive dividends D_1 and D_2
- At the end of the second year the investor sells the share at the price P_2



$$(1): P_0 = \boxed{\frac{D_1}{1+r_e}} + \underbrace{\boxed{\frac{D_2 + P_2}{(1+r_e)^2}}}_{\frac{P_1}{1+r_e}}$$

The pricing formula for a two-year investment horizon equals the stringing together of two one-year horizons

It follows: The price forecast P_2 could be replaced by $\mathbf{P_2 = \frac{D_3+P_3}{1+r_e}}$ $\boldsymbol{with\ P_3 = \frac{D_4+P_4}{1+r_e}}$

III. Investor has a time horizon which is larger than two years

- If the stock price at the end of the period is always substituted by the price the next investor would be willing to pay, it leads to the following general formula which is called the dividend discount model.

$$(1): P_0 = \frac{D_1}{1+r_e} + \frac{D_2}{(1+r_e)^2} + \frac{D_3}{(1+r_e)^3} + \cdots + \frac{D_N}{(1+r_e)^N} + \frac{P_N}{(1+r_e)^N}$$

- If N becomes infinitely large, we get the following formula for the price of a stock. Due to the long investment horizon the present value of the future price P_N will converge to zero.

$$(2): P_0 = \frac{D_1}{1+r_e} + \frac{D_2}{(1+r_e)^2} + \frac{D_3}{(1+r_e)^3} + \cdots + \frac{D_N}{(1+r_e)^N} + \cdots = \sum_{t=1}^{\infty} \frac{D_t}{(1+r_e)^t}$$

Then, the price of a stock equals the present value of all expected future dividend payments.

Pricing of stocks with the dividend discount model

- **Challenge**: How to predict the future dividends?
  - If we want to find the price for the stock today, we would need to predict the dividends until infinity.

- **Different common models for the prediction of future dividends**
  1. Zero-growth model
  2. Constant growth model
  3. Time-varying growth model

Pricing of stocks with the dividend discount model

I. Zero-growth model: Constant dividend payments

- $D_t = D$
- Formula for a perpetual annuity can be applied (constant regular payment at regular points in time)

$$(1): P_0 = \frac{D}{r_e}$$

- Easy scenario, however, not very realistic!

II. Constant growth model: constant dividend growth by factor w

- $D_t = D_{t-1} * (1 + w)$
- Formula for a geometrically growing perpetuity (Gordon growth model)

$$(1): P_0 = \frac{D_1}{r_e - w}, with\ D_1 = D_0 * (1 + w)$$

Pricing of stocks with the dividend discount model

(A.3) The shares of two similar firms should be priced. The model parameters for the firms do not differ except for the growth rate w. Both firms paid dividends of $12.50\ EUR$ per share in the last year, their costs of capital equal 11.8%, and the firm specific growth rates are $w_A = 0\%$ and $w_B = 8\%$. What is the price per share of both firms?

$$(1): P_0^A = \frac{D_1}{r_e - w_A} = \frac{12.50}{0.118 - 0} = 105.93\ EUR$$

$$(2): P_0^B = \frac{D_1}{r_e - w_B} = \frac{D_0 * (1 + w_B)}{r_e - w_B} = \frac{12.5 * (1 + 0.08)}{0.118 - 0.08} = 355.26\ EUR$$

(A.4) A citizen of Zurich holds shares of the local cantonal bank since the bank has paid constant high dividends over the last years. In the last year a dividend of $8\ EUR$ has been paid to the shareholders. For the next year the citizen expects the dividend to increase to $8.50\ EUR$. The current price of the stock equals $215\ EUR$. Please compute the constant growth of the dividend which would justify the current stock price. The costs of capital equal 8%.

$$(1): P_0 = \frac{D_1}{r_e - w}$$

$$(2): P_0 = 215 = \frac{8.50}{0.08 - w}$$

$$(3): w = 0.08 - \frac{8.50}{215} = 4.05\%$$

The dividend discount model with constant dividends or with constantly growing dividends – For which type of firms is it applicable?

- Good usability for big and mature corporations due to…
  - Relatively low risk (small r_e)
  - High dividend ratio
  - Relatively slow growth w
    - A good long-term approximation of firm growth is the country growth since it is, per definition, not possible that a firm always grows with higher growth rates over a long period of time.

- For companies which are not mature and which might act in a dynamic market environment we observe non-constant growth rates. Then, we cannot apply the model of constant or constantly growing dividends. We need to account for the fact that we observe time-varying growth rates.

III. Time-varying growth model (two-phase growth model)

1. Constant growth with factor w until N and then zero-growth

$$(1): P_0 = \underbrace{\boxed{\sum_{t=1}^{N} \frac{D_0 * (1+w)^t}{(1+r_e)^t}}}_{\text{growth } w \text{ (until N)}} + \underbrace{\boxed{\frac{D_0 * (1+w)^N}{r_e}}}_{\text{perpetuity with no growth}} * \frac{1}{(1+r_e)^N}$$

2. Constant growth with factor w_a until N and then constant growth with w_b

$$(1): P_0 = \underbrace{\boxed{\sum_{t=1}^{N} \frac{D_0 * (1+w_a)^t}{(1+r_e)^t}}}_{\text{growth } w_a \text{ (until N)}} + \underbrace{\boxed{\frac{D_0 * (1+w_a)^N * (1+w_b)}{r_e - w_b}}}_{\text{perpetuity with growth } w_b} * \frac{1}{(1+r_e)^N}$$

III. Time-varying growth model (two-phase growth model)

- Both formulas need the dividend of the first year D_1 as an input factor. This information can be given in two ways.
  a. As a direct prediction (analyst prediction) for the following year
  b. As a continued value of the dividends paid in the last period D_0. To derive a prediction for the next year, a dividend growth rate is needed. The general formula for these growth rates and the pricing formulas can be derived as:

$$(1): D_1 = D_0 * (1 + w) \Rightarrow D_1 = D_0 * (1 + w_a)$$

$$(2): P_0 = \sum_{t=1}^{N} \frac{D_0 * (1 + w)^t}{(1 + r_e)^t} + \frac{D_0 * (1 + w)^N}{r_e} * \frac{1}{(1 + r_e)^N}$$

$$(3): P_0 = \sum_{t=1}^{N} \frac{D_0 * (1 + w_a)^t}{(1 + r_e)^t} + \frac{\mathbf{D_0} * (1 + w_a)^N * (1 + w_b)}{r_e - w_b} * \frac{1}{(1 + r_e)^N}$$

(A.5) A stock has paid a dividend of $2\ EUR$ in the last year. It is expected that the dividend will grow at a rate of 8% over the next three years. Afterwards, the growth will go down to a permanent rate of 4%. What is the value of the stock if the costs of capital equal 12%?

$$(1): P_0 = \sum_{t=1}^{N} \frac{D_{\mathbf{0}} * (1 + w_a)^t}{(1 + r_e)^t} + \frac{D_{\mathbf{0}} * (1 + w_a)^N * (1 + w_b)}{r_e - w_b} * \frac{1}{(1 + r_e)^N}$$

$$(2): P_0 = \frac{2 * 1.08^1}{1.12^1} + \frac{2 * 1.08^2}{1.12^2} + \frac{2 * 1.08^3}{1.12^3} + \frac{2 * (1 + 0.08)^3 * (1 + 0.04)}{0.12 - 0.04} * \frac{1}{(1 + 0.12)^3}$$

$$(3): P_0 = 28.89\ EUR$$

(A.6) Analysts predict the dividends for firm A to equal $2.50\ EUR$, $2.75\ EUR$, and $2.96\ EUR$. It is expected that the company will grow after these three years with a constant rate of 4%. The costs of capital equal 12%. The current stock price is $36\ EUR$. Is it a fair price?

$$(1): P_0 = \sum_{t=1}^{N} \frac{D_{\mathbf{0}} * (1+w_a)^t}{(1+r_e)^t} + \frac{D_{\mathbf{0}} * (1+w_a)^N * (1+w_b)}{r_e - w_b} * \frac{1}{(1+r_e)^N}$$

$$(2): P_0 = \frac{2.50}{1.12^1} + \frac{2.75}{1.12^2} + \frac{2.96}{1.12^3} + \frac{2.96 * (1+0.04)}{0.12-0.04} * \frac{1}{(1+0.12)^3}$$

$$(3): P_0 = 33.92\ EUR$$

The stock is currently overvalued.